

APM 1110: FA 3

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<https://github.com/hamiangg/APM-1110---FAs-SAs>

5.

A geospatial analysis system has four sensors supplying images. The percentage of images supplied by each sensor and the percentage of images relevant to a query are shown in the following table.

Percentage of images supplied by each sensor

```
s1 <- 15
s2 <- 20
s3 <- 25
s4 <- 40
```

Percentage of relevant images by each sensor

```
r1 <- 50
r2 <- 60
r3 <- 80
r4 <- 85
```

Overall percentage of relevant images

```
overall_r <- (.15 * .50) + (.20 * .60) +
              (.25 * .80) + (.40 * .85)
```

```
overall_r
```

```
## [1] 0.735
```

```
overall_r * 100
```

```
## [1] 73.5
```

0.735 or 73.5% is the overall percentage of an image being relevant to the query, supplied by the four sensors.

6.

A fair coin is tossed twice.

Probabilities of an event

```
P_E1 <- 2/4  
P_E2 <- 2/4  
P_E3 <- 2/4
```

```
P_E1
```

```
## [1] 0.5
```

```
P_E2
```

```
## [1] 0.5
```

```
P_E3
```

```
## [1] 0.5
```

Testing pairwise independence

```
P_E1_E2 <- 1/4  
P_E1 * P_E2
```

```
## [1] 0.25
```

```
P_E1_E2
```

```
## [1] 0.25
```

```
P_E1_E3 <- 1/4  
P_E1 * P_E3
```

```
## [1] 0.25
```

```
P_E1_E3
```

```
## [1] 0.25
```

```
P_E2_E3 <- 1/4  
P_E2 * P_E3
```

```
## [1] 0.25
```

```
P_E2_E3
```

```
## [1] 0.25
```

The three events are pairwise independent, as the probability of their intersection is 0.25 for each pair, while also getting 0.25 as the product of their individual probabilities. # Testing for mutual independence

```
P_E123 <- 1/4  
P_E1 * P_E2 * P_E3
```

```
## [1] 0.125
```

```
P_E123
```

```
## [1] 0.25
```

Since 0.25 is not equal to 0.125, the three events are not mutually independent.